

冲激响应不变法

and Bilinear, $T = 0.5$

Ex 1. $H(s) = \frac{2}{s^2 + 4s + 3}$

1. $H(s) = \frac{1}{s+1} - \frac{1}{s+3}$, ($z = e^{sT}$)

$$\Rightarrow H(z) = T \cdot \left(\frac{1}{1 - e^{-T} z^{-1}} - \frac{1}{1 - e^{-3T} z^{-1}} \right) \Big|_{T=0.5}$$
$$= \frac{0.19 z^{-1}}{1 - 0.83 z^{-1} + 0.14 z^{-2}}$$

2. $s = \frac{z-1}{T} = 4 \times \frac{1-z^{-1}}{1+z^{-1}}$

$$H(z) = H(s) \Big|_{s=4 \frac{1-z^{-1}}{1+z^{-1}}} = \frac{z(z^{-1}+1)^2}{3z^2 - 2.6z + 1.5}$$